

MouseClaimBench: Integrity-Aware, Non-Compensatory Authorization of Mouse-Brain Claims

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Abstract

Mouse-brain studies combine evidence with distinct meanings. Reproducible execution or accurate prediction may support a bounded result while leaving mechanism, causality, or digital-twin scope unauthorized. MouseClaimBench is an integrity-aware knowledge-based system whose non-compensatory contracts separate domain evidence, structural conformance, provenance integrity, and layer-resolved deficits. Profile v2 links 10 claims, 22 predicate contracts, 60 claim–evidence relations, and 124 observation slots to sources and relation-specific rationales. The profile remains author-defined rather than independently content-validated. Python and a separately implemented Answer Set Programming (ASP) path agree on domain authorization and deficits across 5,677 contract cases. External Shapes Constraint Language (SHACL) independently recovers structural

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conformance and structural deficits for the same packages, while all eight gate combinations satisfy the executable $S \wedge A \wedge I$ rule. A further 55,031 property checks show no violation. Counterfactual tests confirm 9,839 sufficient repairs and 51,480 individual-necessity checks. The integrity gate defines 13 deficit types and blocks 2,550 attacked packages spanning all 255 non-empty compositions of eight original attack families across 10 claims. Twenty internally coherent forgery controls intentionally escape because no external trust anchor is available. Two pre-access frozen Distributed Archives for Neurophysiology Data Integration (DANDI) protocols preserve one availability refusal and one bounded predictive authorization across 32 mice. A dependence-aware Machine Intelligence from Cortical Networks (MICRONS) application confirms only a local observational structure–function association in two hold-outs. These results establish traceable author-policy execution, not biological truth, criterion validity, human utility, expert consensus, or a complete digital mouse brain.

Keywords: Knowledge-Based Systems, Scientific Claim Authorization, Evidence Provenance, Graph Validation, Mouse-Brain Models, Reproducibility

1. Introduction

Computational neuroscience increasingly depends on reusable datasets, predictive models, connectomic reconstructions, simulation software, and derived statistical artifacts. These resources make analyses more powerful, but they also create a difficult knowledge-engineering problem. The evidence that supports one statement is frequently insufficient for another. Held-out

response prediction does not identify the biological topology that generated the response. Repeated execution does not provide independent biological replication. A local structure–function association does not establish causal
10 direction or whole-brain coverage. The technical problem addressed in this work is how to encode these distinctions as an executable authorization policy and how to verify that the resulting system does not silently promote a stronger claim.

Mouse-brain modelling is a demanding application because its public re-
15 sources cover distinct biological scales. Allen tracer experiments describe mesoscale connectivity, population recordings characterize behaviour and neural response, Sensorium benchmarks measure predictive performance, and Machine Intelligence from Cortical Networks (MICRONS) links anatomy and function inside a restricted cortical volume [1–4]. The Virtual Mouse
20 Brain, Brain Modeling ToolKit (BMTK), and detailed primary visual cortex (V1) models already provide established simulation capabilities [5–7]. Digital Twin Brain proposals extend this landscape through assimilation and multiscale modelling [8, 9]. None of these resources should be reduced to a common quality score, and a validation layer should not reimplement their
25 simulators.

Existing scientific claim verification primarily retrieves documentary evidence and predicts whether a passage, table, or figure supports a proposition [10–13]. Recent benchmarks also evaluate whether language models can extract and link claim–evidence pairs from complete articles [14]. As-
30 surance cases and knowledge-based systems provide explicit claims, supporting evidence, argument structures, and inference procedures [15–17]. Prove-

nance and graph validation are already standardized through PROV Ontology (PROV-O) and Shapes Constraint Language (SHACL) [18, 19]. These components are necessary but do not by themselves answer whether a package containing successful prediction, incomplete lineage, overlapping cohorts,
 35 or observational topology evidence authorizes a particular scientific interpretation.

The central design decision of MouseClaimBench is to separate structural conformance, package integrity, and domain authorization. Structural
 40 conformance asks whether the required fields and relations exist. Package integrity asks whether profile identity, content hashes, derivation references, independence declarations, and attestations are internally coherent. Domain authorization asks whether every mandatory evidence block for a bounded claim has passed its declared predicate. A package is authorized only when
 45 all three layers pass. Missing or failed requirements are returned together, so a topology deficit cannot be hidden by predictive performance and an invalid lineage cannot be hidden by an otherwise complete schema.

This paper develops the integrity-aware authorization version of MouseClaimBench. Its central contribution is an authorization contract that binds
 50 four elements that are often evaluated separately: domain evidence semantics, structural admissibility, provenance integrity, and category-complete non-compensatory deficits. The output is deliberately limited to *authorized under profile Π* or *not authorized under profile Π* , accompanied by every violated deficit category and an actionable witness for each detected cate-
 55 gory. This is not a probability that a scientific statement is true. It is a reproducible decision under one named policy whose content remains open

to revision and external criticism.

Research contributions. The paper makes the following contributions:

- A machine-readable knowledge-acquisition record for 10 bounded claims,
60 22 predicate contracts, and all 60 claim–evidence relations. Every relation has a stable identifier, claim-specific necessity rationale, scope, exceptions, sources, and explicit author-policy status.
- An integrity-aware authorization contract that jointly binds evidence-schema admissibility, domain claim requirements, provenance relations,
65 and non-compensatory decisions for computational mouse-brain studies.
- An interoperable Resource Description Framework (RDF)/PROV-O and JSON-LD representation with independently executed SHACL shapes. Python and a separately implemented Answer Set Programming (ASP) path agree on domain decisions and deficits across all 5,677 cases.
70 SHACL independently recovers structural conformance and structural deficits without treating a scientific failure as malformed data.
- A one-edge structural policy perturbation over 221 profiles and 3,094 fixed profile–case evaluations, together with counterfactual tests of explanation sufficiency, minimality, witness coverage, and information retention.
- 75 • An explicit $S \wedge A \wedge I$ final gate with layer-resolved traces, evaluated over all eight logical states, component ablations, 2,550 attacked packages spanning all 255 non-empty compositions of eight original attack families across 10 claims, 100 extended integrity regressions, and 20 coherent-forgery controls that expose the unresolved external-trust boundary.

- 80 • Two pre-access frozen applications to external Distributed Archives for Neurophysiology Data Integration (DANDI)/Neurodata Without Borders (NWB) resources, a post-outcome threshold-sensitivity analysis that does not replace their criteria, and a dependence-aware MICRONS application. Positive and negative outcomes are preserved without changing endpoints, 85 thresholds, or selection rules after numerical access.

The claimed novelty is the evaluated composition of these elements for artifact-grounded scientific authorization. The work does not claim to invent assurance cases, graph validation, provenance ontologies, formal rule checking, or scientific fact verification. It also does not claim a consensus 90 taxonomy, automatic peer review, biological truth, causal identification from observational data, or a complete digital mouse brain. Section 2 positions the contribution, Section 3 defines the knowledge-based method, Section 4 reports the experimental evaluation, and Sections 5 and 6 discuss its scope and conclusions.

95 2. Related Work

MouseClaimBench lies between scientific claim verification, assurance-case reasoning, semantic validation, and neurocomputational model assessment. Table 1 summarizes the closest methodological families. The comparison is capability-based. It does not assert that each family has one uniform 100 implementation.

2.1. Relationship to the preceding MouseBrainBench study

MouseClaimBench follows the MouseBrainBench validation study [23], but it does not present the earlier model evaluations as new experiments.

Table 1: Capability comparison with the closest methodological families. “Partial” denotes a capability that is narrower, implicit, or implementation-dependent.

Family	Decision object	Graph validation	Package integrity	Scientific authorization
Scientific claim verification [10, 11]	Documentary support or contradiction	No	Source traceability	Partial
Assurance cases [17, 20]	Structured argument for a system property	Partial	Partial	Context-specific
SHACL and PROV-O [18, 19]	Graph conformance and provenance interchange	Yes	Partial	No
Model V&V and reproducibility [21, 22]	Credibility, execution, and context of use	Partial	Partial	Partial
Neural model benchmarks [2, 3]	Prediction, simulation, or biological analysis	No	Dataset- specific	One endpoint
MouseClaimBench	Profile-relative authorization from scientific artifacts	Yes	Thirteen declared integrity- deficit types	Ten bounded claim types

MouseBrainBench introduced a claim-aware validation framework, a diag-
105 nostic Mechanistic Identifiability Score, and a non-compensatory gate over
prediction, reproducibility, topology, direction, local structure–function as-
sociation, and cost. It evaluated synthetic systems and bounded Allen, Sen-
sorium, International Brain Laboratory (IBL), and MICRONS cases. The
present study reuses five frozen result packages from that work only as ret-
110 spective inputs to a different authorization problem. Table 2 separates
inherited material from the contribution evaluated here.

The new contribution is therefore not another analysis of the same five

Table 2: Relationship to the preceding MouseBrainBench study. Reused result packages are retrospective inputs and are not counted as new external validation. “New here” identifies components whose implementation and evaluation constitute the present contribution.

Component	MouseBrainBench	MouseClaimBench
Primary question	Which bounded interpretations are supported by model-evaluation evidence?	Does a versioned evidence package authorize one bounded claim under structural, integrity, and domain gates?
Claim representation	Validation dimensions, evidence contracts, diagnostic score, and conjunctive gate	10 claim types, 22 predicate contracts, 60 direct requirements, and category-complete deficits
Reused artifacts	Allen, Static and Dynamic Sensorium, IBL, and MICRONS evaluations	Five frozen packages mapped retrospectively to profile v2
Authorization engine	Bounded claim gate for the original benchmark	Observation admissibility, typed statuses, complete witnesses, and non-compensatory authorization
Standards layer	Not a central evaluated contribution	RDF/PROV-O, JSON-LD, generated SHACL, and graph-isomorphism checks
Integrity layer	Reproducibility and artifact controls within source analyses	Eight relational controls over profile, content, lineage, independence, attestations, and traceability
Independent execution	Original benchmark implementations and controls	Python plus separately implemented ASP, with external SHACL structural validation
Threat evaluation	Not the primary question	Ablations, all 255 declared attack compositions per claim, and coherent-forgery escapes
External protocols	Retrospective public-resource evaluation	Two additional DANDI protocols frozen before numerical access

resources. It is the versioned knowledge profile, admissibility semantics, complete multi-deficit authorization, external graph conformance, relational package-integrity layer, independent rule execution, controlled attack evaluation, and two pre-access DANDI protocols. The dependence-aware MICRONS reanalysis extends uncertainty treatment within the same volume and remains internal reproduction. This separation is also recorded in the repository overlap disclosure and release manifests.

120 2.2. *Scientific claims, knowledge systems, and assurance*

Scientific claim verification commonly decomposes the task into retrieval, evidence selection, and support or contradiction classification [24]. SciFact introduced expert-written claims grounded in scientific abstracts, while MultiVerS and SciFact-Open extended full-document and open-domain reasoning [10, 11, 25]. SciTab, SciVer, SciTrue, PhyVer, and later multimodal approaches include tables, figures, physical measurements, and richer evidence paths [12, 13, 26–28]. CLAIM-BENCH evaluates the extraction and linkage of claim–evidence pairs by language models [14]. These systems address documentary support. MouseClaimBench instead assumes that evidence artifacts and their source predicates have been produced, then evaluates whether their scientific type and provenance satisfy a declared authorization contract. A retrieved sentence cannot substitute for an intervention, an independent cohort, or a topology control.

Rule-based expert systems separate domain knowledge, facts, inference, and explanation [15, 16]. Selective prediction adds a learned reject option when a prediction should not be issued [29]. MouseClaimBench instead returns all typed contract deficits and does not learn an abstention threshold.

Classical validation distinguishes structural correctness from the validity of acquired domain knowledge [30, 31]. This distinction remains central here.
140 Exhaustive tests and multiple solvers can establish that an implementation follows its profile, but they cannot establish that the profile is a complete or universally accepted account of neuroscience.

Assurance cases organize claims, arguments, assumptions, defeaters, and evidence for a defined context of use [17, 20]. Current work also studies
145 assurance cases as attributed graphs and identifies provenance-related structural differences between human- and machine-generated arguments [32]. Compliance-by-construction argument graphs combine typed arguments, deterministic admissibility constraints, and PROV-aligned ledgers for AI-assisted certification records [33]. Claim-level auditability has similarly been
150 proposed for research agents through persistent provenance and protocolized validation [34]. MouseClaimBench does not use a generative model to draft an assurance case. It operationalizes a domain claim profile, tests category-complete deficits across independent execution paths, measures false authorization under controlled package attacks, and evaluates frozen
155 protocols on external scientific artifacts.

2.3. Semantic validation, provenance, and reproducibility

SHACL validates an RDF data graph against a shapes graph and returns a machine-readable validation report [19]. PROV-O represents entities, activities, agents, and derivation relations for interoperable provenance [18].
160 MouseClaimBench reuses both standards. Profile definitions and evidence packages are exported as RDF/JSON-LD, and an external SHACL engine checks field and relation constraints. Consequently, structural graph valida-

tion and provenance interchange are not presented as novel results.

The remaining problem is semantic and relational. A syntactically valid
165 graph may cite the wrong profile, reuse the same artifact as independent
evidence, declare overlapping cohorts as disjoint, contain incompatible attestations, or authorize a claim whose mandatory scientific block failed. These conditions cross the boundary between graph shape, package integrity, and domain policy. MouseClaimBench handles them through an additional non-
170 compensatory integrity gate and returns the complete witnesses for every detected deficit.

Reproducibility guidance, Findable, Accessible, Interoperable, and Reusable (FAIR) principles, model cards, datasheets, and research-software standards improve disclosure and reuse [22, 35–39]. NWB provides a com-
175 mon neurophysiology data language and underlies reuse through DANDI [40]. MouseClaimBench records exact resource versions, asset identifiers, official hashes, code revisions, protocol hashes, and bounded interpretations. It also distinguishes clean computational reproduction, within-resource reproduction, and external study replication. These terms cannot be promoted
180 into one another.

Model V&V and uncertainty-quantification frameworks assess implementation correctness, empirical adequacy, uncertainty, credibility, and context of use [21, 41–43]. MouseClaimBench does not replace those analyses. It consumes their resulting artifacts as typed evidence and prevents a passed
185 validation dimension from authorizing another claim whose required evidence is absent.

2.4. *Validation boundaries in computational mouse-brain research*

Mouse-brain resources differ in anatomical scale and experimental purpose. Allen tracer data and derived connectomes support mesoscale structural analysis [1, 44]. Sensorium and Dynamic Sensorium evaluate neural response prediction under static and video stimuli [2, 3, 45]. Recent work extends predictive generalization and analyzes population-response geometry in visual-cortex digital twins, but prediction remains distinct from unique mechanistic identification [46, 47]. MICRONS provides detailed anatomy and function in visual cortex and supports local structure–function analysis [4, 48]. It is not a measured synaptic connectome of the complete mouse brain.

The Virtual Mouse Brain, BMTK, and detailed V1 models solve simulation and network-construction problems at different scales [5–7]. Observational causal-direction benchmarks show that direction requires assumptions beyond association [49]. Digital Twin Brain and MouseDTB proposals further motivate explicit distinctions among signal prediction, topology, direction, intervention, entity specificity, anatomical coverage, and operational updating [8, 9]. MouseClaimBench does not compete with these simulators. It evaluates which bounded statements their associated evidence packages may authorize.

The targeted review of representative related work did not identify a system combining domain-specific evidence predicates, profile-relative non-compensation, complete multi-deficit traces, separately implemented ASP and external SHACL conformance, package-dependence attacks, prospective external protocols, and mouse-brain artifact applications. This observation

supports differentiation, not global priority or dominance over the representative systems in Table 1. Each component has substantial prior art, and the contribution must be judged on the integration, executable evaluation, and
215 scientific boundaries reported here.

3. Integrity-Aware Knowledge-Based Method

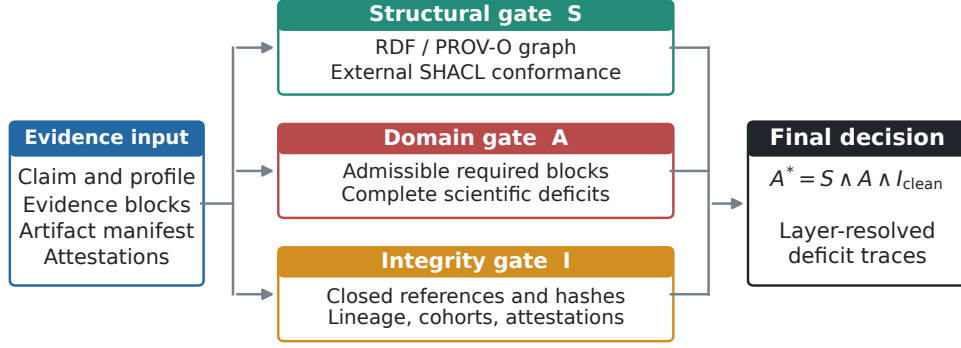
MouseClaimBench receives a versioned authorization profile, a set of evidence blocks, and a manifest of the artifacts from which those blocks were produced. It returns a bounded decision, the complete set of unmet requirements, and a provenance-linked explanation. Fig. 1 shows the complete path.
220 The three validation layers are intentionally separate because a well-formed graph can contain incoherent provenance, and an internally coherent package can still lack the scientific evidence required by a claim.

3.1. Authorization profile and evidence semantics

225 The knowledge profile is represented as

$$\Pi = \langle C, E, Q, O, V, H \rangle, \quad (1)$$

where C is the claim vocabulary, E is the evidence-block vocabulary, $Q(c) \subseteq E$ contains the mandatory blocks for claim c , $O(e)$ contains the observations required when block e declares a pass, V is the profile version, and H is its SHA-256 content identity. The evaluated profile contains 10 claim types, 22
230 evidence-block types, and 60 claim-to-evidence requirements. Table 3 enumerates the claim types, while Fig. 2 shows every required relation. The strict digital-twin claim directly requires 21 blocks. It does not inherit au-



The same claim, evidence blocks, and manifest are evaluated by every gate

Figure 1: Three-gate MouseClaimBench workflow. The same claim, evidence blocks, and manifest determine the RDF/PROV-O representation and enter structural conformance S , domain authorization A , and package integrity I . The gates execute separately and are composed only by $A^* = S \wedge A \wedge I$. A failure at any gate prevents authorization, while the output preserves every violated structural, scientific, and integrity category with an actionable witness.

thorization from the other nine claims and is not a synonym for predictive performance.

235 The profile content follows a recorded acquisition procedure rather than being inferred from the conformance benchmark. The procedure first defines bounded claim wording and prohibited promotions. It then identifies distinct evidence functions from methodological and mouse-brain sources, records scope and exceptions, resolves the internal adversarial audit, and verifies
240 exact executable and bibliographic coverage. The machine-readable basis contains one stable record and one claim-specific necessity rationale for each of the 60 direct relations. It also retains the status *source-grounded author*

Table 3: The 10 exact claim types in profile v2. Required-block counts sum to the 60 claim–evidence relations. Contract cases include independent source, rule, and rationale omissions and sum to 5,677.

Code	Claim type	Blocks	Cases
C1	Bounded predictive performance	3	92
C2	Clean computational reproduction	1	13
C3	Within-resource reproduction	2	46
C4	Independent-study replication	3	90
C5	Topology-specific prediction	5	230
C6	Assumption-conditional direction	4	150
C7	Local observational structure–function association	6	321
C8	Intervention-supported effect	5	228
C9	Directed topology-consistent prediction	10	860
C10	Complete entity-specific mouse-brain digital twin	21	3,647
Total		60	5,677

requirement and *not independently validated* for every relation. The source identifiers justify the methodological function of an evidence block. They are
245 not represented as having prescribed the complete profile.

The separate external-review protocol adapts established content-validity procedures [50, 51]. It requires at least seven independent experts, at least five years of relevant experience, representation of three expertise strata, complete 1–4 ratings, item- and scale-level content-validity indices, binary Fleiss
250 agreement, and resolution of every critical veto. The acceptance thresholds were frozen before recruitment. No eligible rater has completed the protocol, so none of these planned statistics enters the present evidence.

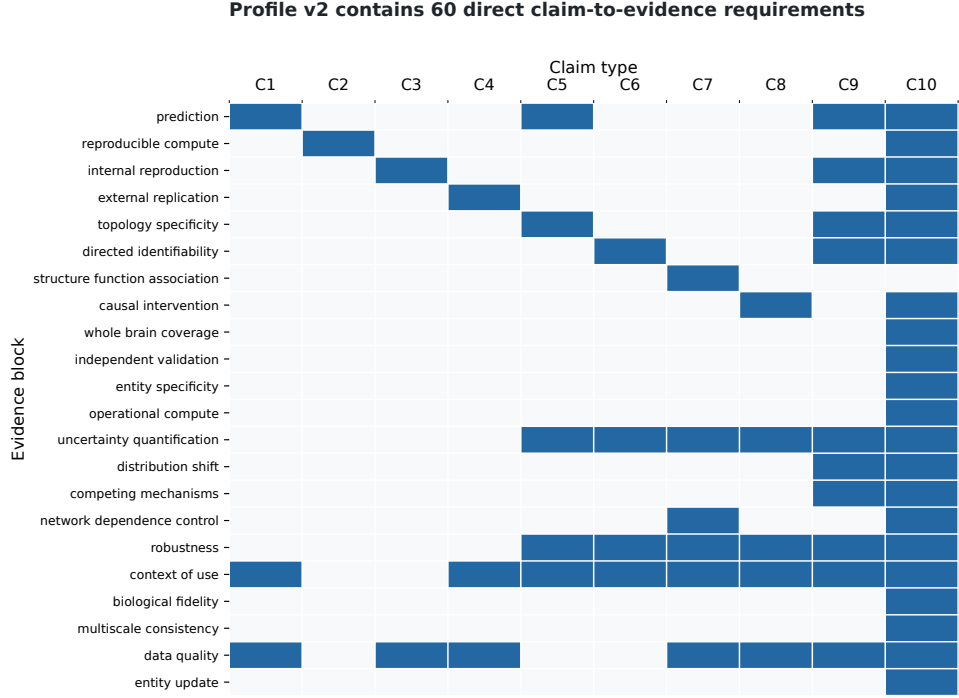


Figure 2: Complete profile-v2 claim–evidence matrix. Columns C1–C10 follow Table 3. A filled cell denotes a mandatory evidence block. The 60 filled cells are direct profile relations. In particular, C10 directly requires 21 blocks and does not inherit authorization from C1–C9.

Predicate ownership is recorded separately from claim authorization. Each of the 22 predicate contracts states whether an upstream source adapter, a release adapter, or the authorization adapter evaluates the sci-
 255 entific or operational rule. The contracts expose 124 required observation slots. Source adapters calculate endpoint-specific statuses and preserve the rule, rationale, observations, and artifact identity. The core authorization engine consumes those attestations, checks their schema and provenance
 260 admissibility, and applies $Q(c)$. It does not impose one universal numerical

threshold across prediction, topology, intervention, anatomical coverage, and reproducibility.

An evidence block is

$$b_e = \langle e, a, x, s, \phi, r \rangle, \quad (2)$$

where $a \in \{P, F, U, N, J\}$ denotes passed, failed, unknown, not targeted, or
 265 requires review, x preserves the original-scale observations, s identifies the
 source artifact, ϕ is the predicate executed by the declared source adapter,
 and r is its rationale. The authorization engine validates the attested predi-
 cate record rather than recomputing every source analysis. It does not nor-
 malize correlation, matched-pair differences, software revisions, interventions,
 270 and anatomical coverage into a shared numerical scale. Let

$$\text{present}(z) = \begin{cases} 1, & z \text{ exists and is non-empty,} \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

A declared pass is admissible only when the block also contains its manda-
 tory observation fields:

$$\text{adm}_\Pi(b_e) = 1[a = P] \text{present}(s) \text{present}(\phi) \text{present}(r) \prod_{o \in O(e)} \text{present}(x_o). \quad (4)$$

Here, $1[\cdot]$ is the indicator function. Consequently, failed, unknown, not tar-
 geted, and requires-review states are all non-admissible. A declared pass is
 275 also non-admissible when its source, predicate, rationale, or a required ob-
 servation is absent. This rule prevents an evidence source from asserting
passed while omitting the target, split, comparator, inferential unit, uncer-
 tainty method, or other observations required by the profile. The complete

scientific deficit set is

$$\Delta_{\Pi}(c, B) = \{e \in Q(c) : \text{adm}_{\Pi}(b_e) = 0\}, \quad (5)$$

280 where an absent block is materialized as unknown. Core authorization is non-compensatory:

$$A_{\Pi}(c, B) = 1[c \in C]1[\Delta_{\Pi}(c, B) = \emptyset]. \quad (6)$$

The engine reports every member of Δ_{Π} with declared and effective states, source, predicate, rationale, observations, and missing fields. It does not select one prioritized explanation or calculate a paper-level score. Category-
 285 complete reporting means that every violated contract dimension is retained without priority masking and has an actionable witness. It does not require exhaustive enumeration of every redundant witness or every distinct cycle within one violation category. Let $T_R(B)$ replace exactly the blocks in a reported repair set R with admissible passing attestations. A category-
 290 complete deficit explanation is counterfactually sufficient and individually necessary when

$$A_{\Pi}(c, T_{\Delta_{\Pi}(c, B)}(B)) = 1 \quad \wedge \quad \forall e \in \Delta_{\Pi}(c, B), \quad A_{\Pi}(c, T_{\Delta_{\Pi}(c, B) \setminus \{e\}}(B)) = 0. \quad (7)$$

This property establishes fidelity to the non-compensatory profile. It does not establish that a human reviewer finds the explanation useful.

3.2. Standards representation and package integrity

295 The profile and evidence package are exported as RDF using PROV-O. Profiles, claims, evidence blocks, artifacts, activities, and decisions receive

stable identifiers. Artifact derivation uses PROV relations, while profile-specific predicates represent mandatory evidence and authorization status. The same graph can be serialized as Turtle or JSON-LD. This layer supports
 300 exchange and inspection. It does not replace the executable profile.

For each profile, MouseClaimBench generates SHACL shapes that require the package and profile identity, all mandatory block records, a status from the allowed vocabulary, source, rule, rationale, and the observations required when a block declares a pass. A third-party `pySHACL` processor validates the
 305 resulting graph and returns structural deficits. The shapes do not require **passed**. A complete block with **failed** status is therefore structurally conformant but scientifically non-admissible. This distinction permits $S_{\Pi} = 1$ and $A_{\Pi} = 0$ for the same package. SHACL provides an external structural baseline. It does not authorize a scientific claim or make schema validation
 310 a new contribution. In this release, the structural contract applies to graphs emitted by the MouseClaimBench serializer from typed evidence-package objects. It is not presented as a complete validator for arbitrary hostile third-party RDF graphs, where duplicated or contradictory singleton values would require additional uniqueness constraints.

315 The package-integrity layer evaluates relations that cannot be reduced to the presence of individual fields. Let M contain the profile identity, content-addressed artifact records, derivation edges, block-to-artifact lineage, evidence attestations, and declared independence relations. Its integrity deficit set $I_{\Pi}(M, B)$ covers the 13 conditions in Table 4. Artifact identifiers and
 320 block-lineage declarations must be unique. All artifact references from derivations, block lineage, attestations, independence pairs, and disjoint-cohort

pairs must resolve. Block lineage and attestations must reference supplied evidence blocks. Every supplied block requires lineage and at least one attestation, while each direct attestation must agree with the represented block status. Independence and cohort-disjointness relations must be irreflexive. The taxonomy also protects contract identity, artifact content, derivation order, and declared independence. These assets follow established concerns in secure provenance, assurance cases, and the avoidance of pseudoreplication [17, 52, 53].

The cycle check uses an iterative depth-first traversal, which avoids recursion limits for large packages. Independence is evaluated from content identity and from the composite identity (`study_id`, `data_generation_id`) when both fields are non-empty. The implementation does not assume that `data_generation_id` is globally unique across studies. These checks establish internal package consistency under the declared metadata. They do not prove that a source is trustworthy or that two laboratories are scientifically independent. Profile identity is intentionally checked at both the structural graph boundary and the manifest-integrity boundary. The structural check prevents a graph from masquerading as an instance of a different executable profile, while the integrity check preserves the same invariant in the package manifest. This defense-in-depth overlap is reported explicitly rather than treated as evidence that the two gates are disjoint.

Structural conformance remains an independently executed gate. The graph is deterministically derived from the same claim, evidence blocks, and manifest evaluated by the other gates:

$$G_{\Pi}(c, B, M) = \text{RDF}_{\Pi}(c, B, M). \quad (8)$$

Table 4: Thirteen-type declared integrity threat model. The original benchmark covers eight attack families individually, pairwise, and in every non-empty composition. Extended regressions cover duplicate identifiers, reflexive relations, dangling references, unknown blocks, status disagreement, and missing attestations. Neither experiment represents attack prevalence or adaptive adversaries.

Attack	Protected asset	Violated invariant
Profile-version substitution	Authorization contract	Identifier, version, and profile hash must match
Duplicate artifact identifier	Package namespace	Every artifact identifier must be unique within a package
Duplicate block lineage	Evidence trace	Every block must have one consolidated lineage declaration
Artifact-hash tampering	Source content	Declared and observed content identities must agree
Dangling provenance reference	Derivation graph	Every referenced artifact must resolve
Circular provenance	Derivation graph	Artifact derivation must be acyclic
Duplicate independent artifact	Independence declaration	Independence pairs must be irreflexive and cannot share content or composite generation identity
Overlapping independent cohorts	Inferential units	Cohort-disjointness pairs must be irreflexive and cannot share subjects
Contradictory attestation	Evidence status	One block cannot have incompatible source attestations
Missing block lineage	Evidence trace	Every supplied block must resolve to at least one source artifact
Unknown block reference	Evidence trace	Lineage and attestations must reference supplied blocks
Block-attestation mismatch	Evidence status	Every direct attestation must agree with the represented block status
Missing block attestation	Evidence trace	Every supplied block must have at least one source attestation

Let $S_{\Pi}(G) = 1$ exactly when external SHACL validation reports no structural violation against the shapes generated from Π . The field checks in S_{Π} intentionally overlap with defensive admissibility checks in the Python core, but S_{Π} is not absorbed into A_{Π} . Final authorization requires structural
 350 conformance, a complete scientific contract, and an integrity-clean package:

$$A_{\Pi}^*(c, B, M) = S_{\Pi}(G_{\Pi}(c, B, M))A_{\Pi}(c, B)1[I_{\Pi}(M, B) = \emptyset]. \quad (9)$$

No control can compensate for another. A dedicated final authorization object executes this equation and preserves separate structural, domain, and integrity traces. Every reported deficit retains its corresponding witness.

3.3. Independent execution, domain-gate property checks, and release control

355 The authorization semantics are implemented through three execution paths with deliberately different scopes. Source-specific Python adapters execute the scientific predicates and construct attested evidence blocks. The primary Python engine evaluates field presence, observation admissibility, domain requirements, and detailed scientific explanations. The external
 360 SHACL path evaluates only the structural contract encoded in the RDF graph. A separately implemented ASP program receives declared statuses, metadata-presence facts, and observation-presence facts. It derives effective states, authorization, and deficits within Clingo for every generated domain-contract case. Relational package integrity is evaluated by its own Python
 365 component before the three gate results are composed. The paths share the profile specification but not the same inference code. Agreement between Python and ASP, and conformance of SHACL to structural expectations,

provide implementation evidence only within their declared scopes. They do not validate the shared scientific policy.

370 The domain-gate property contract contains six profile-relative properties of A_{Π} . Soundness requires every domain-authorized claim to have all required admissible blocks. Completeness requires every package with all required admissible blocks to be domain-authorized. Deficit identity requires the reported set to equal the set defined in Eq. 5. Monotonic degradation re-
375 quires replacing a passed block with a failed block to preserve or enlarge the deficit set and never create domain authorization. Domain decisions must also be invariant to evidence input order and irrelevant unprofiled blocks. Outside-profile closure is checked separately by requiring every claim outside C to remain outside the authorization boundary. Irrelevant-evidence
380 invariance does not apply to arbitrary modifications of a final evidence package. Every supplied block must still satisfy manifest lineage and attestation requirements, even when it is irrelevant to $Q(c)$. These are formalized properties tested over generated packages, not an exhaustive proof over arbitrary profiles and not statements about biological truth.

385 Release artifacts record protocol hashes, profile hashes, source revisions, exact public asset identifiers, and official data hashes. A release is rejected when a required artifact is absent, any source revision ends in **-dirty**, an expected benchmark decision changes, or a broad prohibited claim becomes true. The release gate therefore checks reproducibility and scope together. It
390 cannot guarantee journal acceptance, profile consensus, or human usefulness of an explanation.

4. Experimental Evaluation

The evaluation is organized into four evidence classes. Knowledge acquisition asks whether the author-defined policy is fully traceable. Specification verification asks whether the implementation follows that policy and whether its explanations are counterfactually faithful. Controlled threat evaluation asks which declared integrity violations are detected and which coherent forgeries remain outside the trust model. Empirical application asks whether bounded decisions are preserved on existing and pre-access frozen mouse-brain resources. Performance is reported separately and is not used as scientific validation. Table 5 defines the evidence unit and maximum interpretation for every question.

4.1. Experimental design and decision rules

The acquisition audit compares the executable profile with its machine-readable basis and bibliography. Acceptance requires exact coverage of 10 claim definitions, 22 evidence blocks and predicate contracts, and 60 claim-evidence relations. Every relation must have a unique identifier and claim-specific necessity rationale. Every predicate must state evaluation ownership, decision-rule scope, required observations, sources, and exceptions. All source identifiers must resolve to the bibliography, and the audit must retain false values for independent content validity and human validation. The historical 29-item v1 panel packet is excluded. A separate v2 protocol contains 92 prospective review items covering the 10 claims, 22 predicates, and 60 relations. It requires at least seven eligible independent experts and predefined item content-validity, scale content-validity, agreement, and critical-veto con-

Table 5: Evaluation layers, evidence units, and admissible interpretations. Knowledge acquisition, specification verification, controlled threat evaluation, and empirical application remain separate so that software conformance is not reported as scientific content validation.

ID	Question	Evidence unit	Maximum interpretation
K1	Is the scientific policy traceable?	10 claims, 22 predicates, 60 relations, 124 observation slots, and bibliography resolution	Complete author-policy acquisition record, not independent content validity
V1	Are the three gates and their deficits implemented exactly?	5,677 Python/ASP domain cases, 5,677 SHACL structural cases, five JSON-LD round trips, eight gate states, and 10,000 random domain packages	Profile-relative semantic, structural, interoperability, and compositional conformance
V2	How do one-edge changes and repairs affect fixed decisions?	221 profiles on 14 fixed decisions and 10,000 counterfactual packages	Monotonicity and explanation fidelity relative to v2
T1	Does integrity block declared package attacks?	360 single/pairwise attacked packages, 2,550 attacked packages spanning 255 non-empty compositions, 100 extended regression packages, ablations, and 20 trust-boundary controls	Resistance to declared invariants with explicit coherent-forgery escapes
P1	Is the implementation usable at larger scale?	Up to 10,000 packages and 5,000 artifacts per package	Descriptive performance on one host
A1	Does the profile preserve bounded mouse-brain interpretations?	Five target claims, five strict twin claims, and dependence-aware MICRONS inference	Artifact-grounded authorization under profile v2
A2	Can a frozen contract retain unseen positive and negative outcomes?	Two versioned DANDI resources with pre-access protocols	Prospective workflow evidence within each resource

ditions. It has zero completed independent raters and is not counted as evidence.

The contract benchmark generates eight case families from every profile claim. They contain pristine complete packages, one changed status, one
420 omitted mandatory block, one omitted required observation, independent omission of source, rule, or rationale, and all pairwise combinations of the four non-passing states. For a claim with q_c required blocks and m_c mandatory observation fields across those blocks, the number of cases is

$$N_c = 1 + 8q_c + m_c + 16 \binom{q_c}{2}. \quad (10)$$

The terms represent one pristine package, four status mutations, one block
425 omission, and three metadata omissions per block, one omission per observation field, and 16 status combinations per block pair. Table 3 reports every N_c , which sums to 5,677. Each case has an exact expected domain authorization and deficit set defined from Eqs. 4–6. Python and the separately implemented ASP program evaluate all 5,677 domain cases. SHACL evalu-
430 ates structural conformance and structural deficits over the same packages. Non-passing statuses remain structurally valid, while block, metadata, and required-observation omissions are structural failures. JSON-LD round-trip equivalence is evaluated through graph isomorphism. A separate truth-table benchmark evaluates all eight combinations of S_Π , A_Π , and integrity confor-
435 mance against Eq. 9.

Domain-gate property-based verification uses 1,000 deterministic random evidence packages for each of the 10 claims. Every package is evaluated for soundness, completeness, exact deficits, input-order invariance, irrelevant-evidence invariance, and monotonic degradation whenever a passed block

exists. Outside-profile closure is evaluated separately. Property-based tests generate additional status and observation mutations during the software suite. The reported 55,031 checks refer to the frozen deterministic experiment over A_Π , not to an exhaustive proof over arbitrary future profiles or complete package-integrity behaviour.

The one-edge structural policy perturbation holds 14 artifact instances fixed: the bounded target and strict complete-twin claim for each of five retrospective cases and two pre-access DANDI cases. It evaluates the baseline profile, all 60 profiles obtained by removing one required relation, and all 160 profiles obtained by adding one currently unrequired evidence block to one claim. Let $x_j = (c_j, B_j)$ denote one fixed claim and evidence package. For variant Π_k , the flip indicator is

$$F_{jk} = 1[A_{\Pi_k}(c_j, B_j) \neq A_\Pi(c_j, B_j)]. \quad (11)$$

Core authorization is used deliberately to isolate the scientific relation policy while manifests remain pristine and graphs remain structurally conformant. Removing a relation is a weakening perturbation and can only expand authorization. Adding a relation is a conservative extension and can only contract it when the added block is not admissibly passed. The resulting flips are therefore expected monotonicity probes under one-edge changes. They identify decisive relations in the fixed packages, but do not characterize broad policy dependence, scientific plausibility, or alternative expert profiles.

Counterfactual explanation fidelity is evaluated on another deterministic set of 1,000 packages per claim. Every non-authorized package is repaired by replacing exactly the reported deficits with complete passing attestations. The repaired package must authorize with no residual deficit. Each repaired

block is then restored individually to its original defective state. Authoriza-
 465 tion must again fail with that block as the sole remaining deficit, implement-
 ing Eq. 7. Every witness must retain a non-empty block name, source, rule,
 and rationale. Finally, each of the 60 pristine claim–block relations is de-
 graded independently. A single-reason baseline is evaluated descriptively by
 counting the simultaneous deficits it would omit. No claim about human
 470 usefulness is tested.

The original integrity benchmark applies the first eight attacks defined in
 Section 3.2. For each of the 10 claims, it evaluates one pristine package, every
 single attack, and all 28 pairwise attack combinations. This produces 370
 packages, of which 360 are attacked. The benchmark intentionally excludes
 475 the structural gate to isolate the marginal contribution of package-integrity
 controls, including manifest-level profile identity. For integrity configuration
 k , define the domain-plus-integrity decision

$$A_{\Pi,k}^I(c, B, M) = A_{\Pi}(c, B)1[I_{\Pi,k}(M, B) = \emptyset]. \quad (12)$$

Let $\mathcal{D}_{\text{attack}}$ denote the benchmark cases, with claim c_z , blocks B_z , and mani-
 fest M_z for case z . The configuration-specific attack false-authorization rate
 480 is

$$\text{FAR}_{\text{attack}}^{(k)} = \frac{\sum_{z \in \mathcal{D}_{\text{attack}}} 1[z \text{ attacked}] A_{\Pi,k}^I(c_z, B_z, M_z)}{\sum_{z \in \mathcal{D}_{\text{attack}}} 1[z \text{ attacked}]}. \quad (13)$$

We compare the complete domain-plus-integrity configuration with the core
 profile alone, a hash-only baseline, and eight leave-one-control-out configu-
 rations. Exact trace recovery requires the detected deficit codes to equal
 every invariant violated by the applied attacks. Full $S_{\Pi} \wedge A_{\Pi} \wedge I_{\Pi}$ com-
 485 position is evaluated separately by the eight-state truth-table experiment.

Consequently, Table 8 measures failures of integrity configurations under the declared attack model rather than failures of the complete final authorization object. A compositional stress test then evaluates all $2^8 - 1 = 255$ non-empty attack subsets for each claim, producing 2,550 attacked packages.

490 Two further negative controls per claim replace content and its declared hash coherently or forge internally consistent independence metadata. Because these controls contain no contradiction against an external truth anchor, authorization is the expected result. They expose the need for trusted archives, signatures, or external identity evidence rather than being counted as failures

495 under the declared eight-attack model. An extended regression suite adds one pristine package and nine targeted edge cases per claim. Its 100 packages cover dangling attestation and pair references, unknown attested blocks, both directions of block-attestation status disagreement, and missing attestations. It is kept separate from the historical composition benchmark to

500 avoid an uninformative combinatorial expansion. None of these benchmarks is an adaptive security guarantee.

The core authorization-engine microbenchmark is a component test measured on one Apple arm64 host with Python 3.12.13. Pristine authorization batches contain 100, 1,000, and 10,000 packages drawn cyclically from all profile

505 file claims. Each batch is repeated five times after 100 warm-up decisions. A separate timed component expands one valid digital-twin package to 25, 100, 1,000, and 5,000 uniquely content-addressed artifacts and evaluates its local integrity gate. Median wall-clock time and all repetitions are stored. Downloads, source-predicate execution, graph serialization, and external SHACL

510 validation are excluded. These are component microbenchmarks on one im-

plementation and host, not end-to-end scalability or a portable latency guarantee.

Three artifact-evaluation components complement the controlled benchmarks. The first reuses five result packages assembled during the preceding
515 MouseBrainBench study [23] and asks a different question: whether profile v2 authorizes their bounded target claim and strict digital-twin claim. These mappings are retrospective and are not counted as prospective validation. The second applies dependence-aware inference to MICRONS. Its endpoint was fixed after discovery and evaluated in two non-overlapping hold-
520 out windows within one tissue volume. Directed pair regression controls log soma distance, its square, pre- and postsynaptic degree, and coarse cell type. Dependence is addressed with directed dyadic cluster covariance and 1,000 simultaneous node-label Freedman–Lane permutations. These hold-outs provide internal reproduction, not independent biological replication. The third
525 component evaluates the two external DANDI protocols described below.

For the MICRONS permutation test, the full model regresses readout-location similarity y_{ij} on the directed connection indicator c_{ij} and the fixed control vector z_{ij} :

$$y_{ij} = \beta_0 + \beta_1 c_{ij} + z_{ij}^\top \gamma + \varepsilon_{ij}. \quad (14)$$

The reduced model omits c_{ij} . Its residuals are placed in a directed neuron-
530 by-neuron array. The connection indicator is residualized against the same controls, and the residualized coefficient is the test statistic. Each permutation draws one permutation of neuron identifiers and applies it simultaneously to sender and receiver indices of the reduced-model residual array. This preserves reciprocal array structure and the incidence pattern induced by shared

neurons under relabelling. The discovery window fixes the positive direction
 and the analysis specification. Its one-sided permutation value is reported
 descriptively and has no confirmatory interpretation. The dyadic regression
 is reported with a two-sided test and 95% interval in every window. The con-
 firmatory one-sided permutation test is subsequently evaluated in the two
 pre-fixed hold-outs. Its finite-permutation value is $(1 + K)/(1 + R)$, where
 K is the number of permuted coefficients at least as large as the observed
 positive coefficient and $R = 1000$. Discovery and the two hold-outs use re-
 producible streams with seeds 2026080201, 2026080202, and 2026080203. A
 hold-out passes only when $\hat{\beta}_1 > 0$, $p_{\text{dyadic}} \leq 0.05$, and $p_{\text{perm}} \leq 0.05$. Con-
 firmation requires this conjunction in both fixed hold-outs after a positive
 discovery direction. This procedure assumes exchangeability of the reduced-
 model residual array under simultaneous node relabelling.

Two external DANDI protocols were committed before numerical asset
 access. The availability protocol was frozen in commit **b59a497** and the
 predictive protocol in **d039e66**. Their SHA-256 identities are retained in
 the result artifact, and the numerical result was committed later. This Git
 history is temporal evidence for the declared ordering, not third-party prereg-
 istration. DANDI:001176 was selected from metadata describing 29 subjects
 and simultaneous cholinergic axon, cortical acetylcholine (ACh), pupil, and
 treadmill signals [54]. The protocol required at least 20 usable subjects and
 prohibited threshold or endpoint repair. After catalog inspection, 28 schema-
 valid sessions came from only 5 subjects. The predictive model was therefore
 not executed.

DANDI:000039 contains contrast-tuning recordings associated with a pub-

560 lished visual-cortex study [55]. Before inspecting its asset catalog or values,
 the protocol fixed one lexicographically first `behavior+ophys.nwb` asset per
 subject, a trial-level population mean event response, Ridge regression with
 $\alpha = 1$, and an intercept comparator. For contrast q and direction θ in degrees,
 the exact five regressors are $\log(1 + 100q)$, $\sin(\theta)$, $\cos(\theta)$, $\log(1 + 100q) \sin(\theta)$,
 565 and $\log(1 + 100q) \cos(\theta)$. Angular functions are evaluated after conversion
 to radians. The first 60% of trials train the model, the next 20% remain
 reserved, and the final 20% form the test partition. The intermediate parti-
 tion is deliberately unused. It does not affect model fitting, model selection,
 threshold definition, or final evaluation. The `StandardScaler` is fitted on the
 570 training partition only and then applied unchanged to the held-out partition.
 Ridge $\alpha = 1$ is fixed and no tuning occurs after asset access. The per-mouse
 held-out metric is Pearson correlation. Authorization requires at least 25
 subjects, median held-out correlation at least 0.10, a positive 95% mouse-
 bootstrap lower bound, at least 65% positive subjects, and trial-pooled mean
 575 squared error below the intercept. For M mice with n_m held-out trials, this
 metric is

$$\text{MSE}_{\text{pool}} = \frac{\sum_{m=1}^M \sum_{t=1}^{n_m} (y_{mt} - \hat{y}_{mt})^2}{\sum_{m=1}^M n_m}. \quad (15)$$

The implementation stores and compares the two squared-error numerators
 because their denominator is identical. The resulting metric weights mice in
 proportion to their number of held-out trials. All conditions must pass. The
 580 bootstrap uses protocol seed 2026080202 and resamples mice with replace-
 ment 5,000 times. Each replicate calculates the median mouse-level Pearson
 correlation. The reported lower bound is its 2.5th percentile. It therefore
 targets population stability across mice rather than trial-level independence.

These thresholds are author-defined operational acceptance criteria rather
585 than criterion-validated biological standards. Twenty-five mice was intended
to preserve a multi-mouse endpoint and a meaningful mouse-level bootstrap,
but it was not obtained from a formal power calculation. The 0.10 median
correlation requires a non-trivial predictive signal, the positive lower bound
requires population-level sign stability, the 65% rule requires a majority of
590 positive mice while tolerating heterogeneity, and the intercept comparison
requires incremental predictive value. The 20-subject availability boundary
for DANDI:001176 similarly protects the intended cross-subject design but
is not a sufficiency standard. A post-outcome sensitivity analysis changes
one threshold at a time while holding the other frozen conditions fixed. It
595 exposes decision boundaries and cannot retrospectively calibrate or replace
either protocol.

4.2. Knowledge traceability, one-edge policy perturbation, and explanation fidelity

The acquisition audit exactly covers the 10 claims, 22 evidence blocks,
600 22 predicate contracts, 60 claim–evidence relations, and 124 required obser-
vation slots. All 60 relation identifiers and necessity rationales are unique,
and all 20 source identifiers resolve to the bibliography. The ledger separates
upstream predicate execution from authorization checks and marks every
relation as a source-grounded author requirement without independent con-
605 sensus. This satisfies traceability of the current policy. It does not establish
that the policy is complete or optimal.

The one-edge analysis evaluates 221 profiles and 3,094 profile–case com-
binations. The baseline authorizes four of the 14 fixed decisions. Among the

60 weakening variants, only removal of the `data_quality` requirement from
610 bounded prediction changes an authorization. It converts the Dynamic Sen-
sorium target from a documented quality refusal into an authorization. This
isolates the requirement that is decisive for that case rather than showing
that it should be removed. Among 160 conservative extensions, 52 variants
change at least one decision and produce 71 contractions. These contractions
615 follow the non-compensatory monotonicity of the system because an other-
wise authorized case has no admissible attestation for the newly added block.
Ninety-six variants change at least one deficit set even when authorization
does not change. The counts are implementation context, not independent
scientific evidence for the added relations.

Table 6: One-edge structural policy perturbation across 14 fixed artifact decisions. Ex-
pansions are refusals authorized after removing one requirement. Contractions are autho-
rizations withdrawn after adding one block. The monotonicity probe identifies decisive
relations in these packages. The variants are not scientifically validated candidate profiles.

Policy set	Variants	Runs	With flips	Flips	Interpretation
Baseline profile	1	14	—	0	Four bounded authorizations
Remove one required relation	60	840	1	1 expansion	Removing data quality authorizes the otherwise refused Dynamic Sensorium target
Add one unrequired block	160	2,240	52	71 contractions	Positive cases lose authorization when the added block lacks an admissible attestation
Total	221	3,094	53	72	One-at-a-time structural policy envelope

620 Counterfactual fidelity is evaluated on 10,000 packages. Of the 9,839

non-authorized packages, repairing every reported deficit is sufficient for authorization with no residual deficit. Restoring each deficit individually produces 51,480 minimality checks, all of which refuse authorization with the exact retained deficit. The same 51,480 deficit records retain non-empty
625 witnesses, and all 60 pristine single-block degradations return the expected sole deficit. There are 8,632 packages with simultaneous deficits. A single-reason trace would retain 9,839 deficits and therefore omit the remaining $51,480 - 9,839 = 41,641$. This is arithmetic information retention under the generated packages, not independent evidence about explanation quality or
630 human decisions.

4.3. Domain-gate conformance and property-based verification results

Table 7 reports the principal conformance results. The Python engine produces no false authorization or false rejection in 5,677 cases and recovers every expected domain deficit set. The independent ASP implementation
635 matches Python domain status and deficits in all 5,677 cases while deriving admissibility from declared field-presence facts inside Clingo. The SHACL engine classifies the same 5,677 packages against their structural expectations and returns every expected structural deficit. In particular, status-only failures remain structurally conformant and scientifically unauthorized. All
640 eight logical combinations of structural conformance, domain authorization, and integrity conformance produce the final decision required by Eq. 9. The exported profile contains 318 RDF triples representing all 10 claims and 22 evidence blocks. JSON-LD serialization and parsing preserve graph isomorphism in five representative packages: a pristine package, a scientifically
645 failed but structurally conformant package, a structurally incomplete pack-

age, a rich manifest with multiple attestations, and a larger digital-twin-style package with declared independence and cohort-disjointness relations.

Table 7: Conformance and domain-gate property results. Exactness is relative to the frozen profile-v2 contract, and each engine is reported only for its evaluated comparison set.

Evaluation	Cases/checks	Errors	Result
Python contract mutations	5,677	0	Exact domain authorization and deficits
Independent ASP conformance	5,677	0	100% domain-status and deficit agreement
External SHACL conformance	5,677	0	Exact structural conformance and structural deficits
Three-gate composition	8 states	0	Exact $S \wedge A \wedge I$ truth table
JSON-LD round trips	5 graphs	0	Isomorphic RDF graphs
Domain-gate property benchmark	55,031	0	Six core properties and outside closure hold

These results separate two conclusions. Python and ASP agree on domain authorization and scientific deficits. External SHACL validation agrees with
650 the independently declared structural expectations rather than with domain decisions. It does not recompute source predicates or relational integrity. Structural completeness and structural deficit reporting are therefore not unique capabilities of MouseClaimBench. The additional contribution must be assessed through the explicit non-compensatory composition, domain re-
655 quirements, package integrity, controlled attacks, and real applications.

The 10,000 deterministic random packages produce 55,031 domain-gate

property checks with no violation. Soundness, completeness, deficit identity, input-order invariance, and irrelevant-evidence invariance each pass 10,000 checks. Monotonic degradation passes all 5,030 applicable checks. Outside-
660 profile closure also passes. These results establish the operational semantics of A_{Π} for profile v2. They do not establish final-package invariance or prove that the evidence requirements are scientifically complete.

4.4. Integrity attacks, ablation, and component microbenchmarks

Under the declared deterministic threat model, the complete domain-plus-
665 integrity configuration returns the exact integrity-deficit trace in all 370 packages. It produces no false rejection on the 10 pristine packages and no false authorization on the 360 attacked packages. The core profile does not inspect package integrity and therefore authorizes all 360 attacked packages. Hash checking alone blocks content tampering but authorizes 280 packages
670 affected by profile substitution, broken or cyclic lineage, false independence, cohort overlap, contradictory attestations, or missing block provenance. The structural gate is excluded from these counts, so manifest-level profile substitution remains visible instead of being rejected earlier by structural identity matching.

675 The leave-one-out result distinguishes unique necessity from defensive overlap. Removing any of seven original controls creates 10 false authorizations, corresponding to the single-attack package for that omitted control in each claim. Removing only the contradictory-attestation check creates none because the same package also violates block-attestation status consistency.
680 Pairwise attacks containing another enabled control remain blocked. Table 8 gives the exact values, while Fig. 3 shows the contrast between incomplete

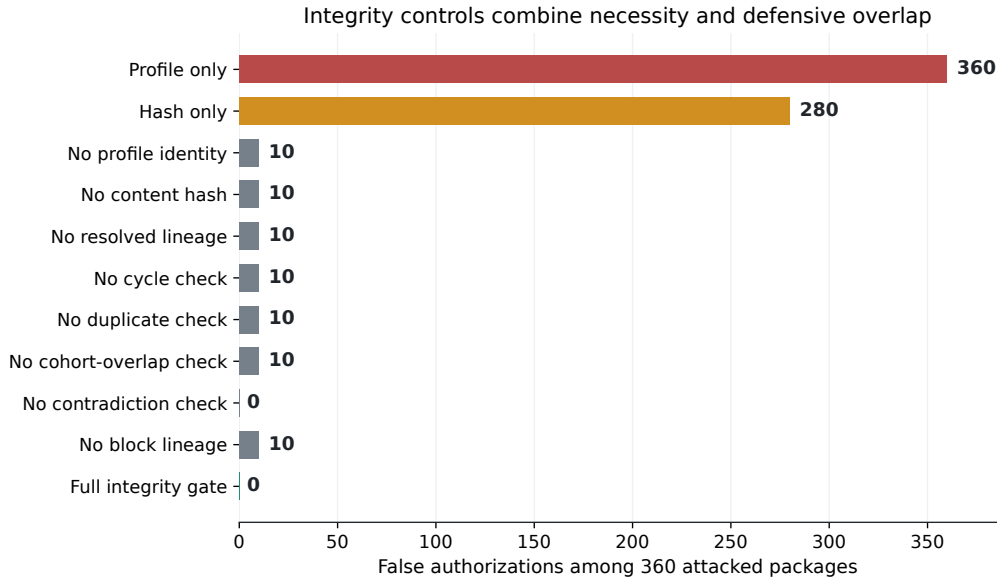


Figure 3: False authorizations among 360 attacked packages across 10 profile claims, where lower values are better. The structural gate is excluded to isolate the domain-plus-integrity configurations. The profile-only baseline authorizes all 360 attacked packages. The hash-only baseline authorizes 280 packages involving identity, lineage, dependence, conflict, or provenance. The full integrity configuration blocks every declared attack. Seven original control omissions expose 10 packages each. Removing the contradiction check alone exposes none because block-attestation status consistency detects the same contradictory packages.

baselines and the complete domain-plus-integrity configuration. The result demonstrates conditional necessity and one explicit redundancy under the declared attack construction. It does not imply sufficiency for every real provenance threat.

The compositional stress test evaluates 2,560 packages, comprising 10 pristine packages and 2,550 attacked packages. For each of the 10 claims, the latter span all 255 non-empty subsets of the eight declared attacks. The

Table 8: False authorizations and false rejections for domain-plus-integrity configurations in the 370-case integrity benchmark. The structural gate is excluded to isolate manifest-level controls. Seven original controls are uniquely necessary for their constructed single-attack families. Removing the contradiction check alone produces no false authorization because block-attestation status consistency independently detects the same contradictory package.

Integrity configuration	False authorizations / 360	False rejections / 10
Profile without integrity	360	0
Hash-only integrity	280	0
Full integrity gate	0	0
Without profile identity	10	0
Without content hash	10	0
Without resolved lineage	10	0
Without cycle detection	10	0
Without duplicate-independence check	10	0
Without cohort-overlap check	10	0
Without contradictory-attestation check	0	0
Without block lineage	10	0

complete domain-plus-integrity configuration produces no false authorization and returns every invariant deficit implied by each attack composition. This removes the original restriction to single and pairwise compositions but does not add new protected invariants. The extended regression suite then evaluates 100 packages and returns exact traces in all cases, with no false authorization among 90 attacked packages and no false rejection among 10 pristine

695 packages. Six additional direct API regressions cover duplicate artifact identifiers, duplicate block-lineage declarations, reflexive independence and cohort relations, and both outcomes of the composite study and data-generation identity rule. The duplicate and reflexive packages return structured final refusals rather than exceptions. These six cases are software regressions and
700 are not added to the frozen 100-package benchmark. The trust-boundary controls provide the necessary negative result. All 20 internally coherent content or independence forgeries remain authorized with no integrity deficit because the package contains no externally verifiable contradiction. The system therefore detects inconsistency with its declared invariants. It does not
705 authenticate the truth of self-consistent source metadata.

Batch throughput ranges from 27,653 to 52,816 authorizations per second. Median time is 0.00189 s for 100 packages, 0.03253 s for 1,000, and 0.36163 s for 10,000. A valid package requires 0.149 ms with 25 artifacts, 0.240 ms with 100, and 5.328 ms with 5,000. These measurements describe the observed
710 range on one host and are not an asymptotic complexity result. All pristine decisions remain authorized at every scale.

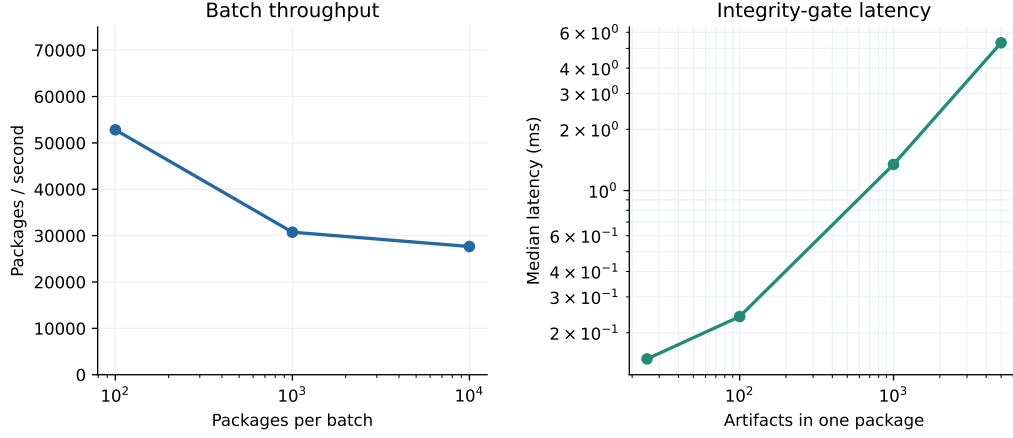


Figure 4: MouseClaimBench component microbenchmarks on one Apple arm64 host using medians from five repetitions after warm-up. The left panel times local core authorization for batches of 100, 1,000, and 10,000 pristine packages. The right panel times the local integrity gate for one valid package containing 25, 100, 1,000, or 5,000 artifacts. Downloads, source predicates, graph serialization, and external SHACL execution are excluded. The measurements are not end-to-end throughput.

Table 9: MouseClaimBench component microbenchmarks on one Apple arm64 host. Every value is the median of five repetitions after warm-up. Downloads, source predicates, graph serialization, and external SHACL execution are excluded.

Workload	Size	Median time (ms)	Throughput
Evidence-package batch	100 packages	1.893	52,816 packages/s
Evidence-package batch	1,000 packages	32.529	30,742 packages/s
Evidence-package batch	10,000 packages	361.629	27,653 packages/s
Single digital-twin package	25 artifacts	0.149	Not pooled
Single digital-twin package	100 artifacts	0.240	Not pooled
Single digital-twin package	1,000 artifacts	1.344	Not pooled
Single digital-twin package	5,000 artifacts	5.328	Not pooled

4.5. Mouse-brain artifacts and prospective external applications

The retrospective profile-v2 mapping contains five target claims and the corresponding five strict digital-twin claims. Three bounded target claims
715 are authorized. Static Sensorium authorizes prediction for its declared five-dataset comparison. The IBL case authorizes topology-specific behavioural prediction under its locked comparison. MICRONS authorizes one local observational structure–function association. Allen directed topology remains unauthorized because prediction, topology, direction, uncertainty, robust-
720 ness, and data-quality requirements do not all pass. Dynamic Sensorium remains unauthorized because the stored response-reliability quality block requires review. None of the five cases authorizes the strict digital-twin claim.

The dependence-aware MICRONS analysis changes the status of the local
725 association without changing its scope. Table 10 reports cohort size, pair counts, effect estimates, uncertainty, and both inferential checks. No permuted statistic is at least as large as the observed positive statistic in any window. With 1,000 permutations, the finite-permutation corrected value is therefore $p = 1/1001 \approx 0.001$ in each window. The discovery value is
730 descriptive because that window selected the positive direction. Both prefixed hold-outs independently pass the positive coefficient, two-sided dyadic, and one-sided permutation criteria. Shared-neuron dependence is addressed more conservatively than naive pair-level inference. All windows nevertheless come from one MICRONS volume, and readout-location similarity is a
735 derived functional descriptor rather than a causal measurement.

The two prospective DANDI outcomes are deliberately asymmetric.

Table 10: Dependence-aware MICRONS inference for the fixed all-pairs/readout-location endpoint. Candidate and connected counts refer to directed pairs. Confidence intervals use the same unit-based degrees of freedom as the directed dyadic test. Discovery fixes the positive direction, so its one-sided permutation value is descriptive. Confirmation requires $\hat{\beta} > 0$, two-sided $p_{\text{dyadic}} \leq 0.05$, and one-sided $p_{\text{perm}} \leq 0.05$ in both pre-fixed hold-outs.

Cohort	Units	Pairs (connected)	$\hat{\beta}$ (SE)	95% CI	p_d	p_p
Discovery	1,000	999,000 (2,095)	0.0106 (0.00200)	[0.00668, 0.01451]	1.39×10^{-7}	0.0010
Hold-out 1	992	983,072 (1,926)	0.0178 (0.00209)	[0.01372, 0.02192]	5.78×10^{-17}	0.0010
Hold-out 2	999	997,002 (1,922)	0.0117 (0.00205)	[0.00763, 0.01569]	1.76×10^{-8}	0.0010

DANDI:001176 fails the frozen availability condition with 5 usable subjects against a minimum of 20. No predictive fit is reported, and the endpoint is not weakened after seeing the catalog. DANDI:000039 yields 32 selected and
740 usable mice. Its median held-out correlation is 0.310, the mouse-bootstrap 95% interval is [0.207, 0.369], and 93.75% of subjects have positive correlation. The stored model and intercept squared-error sums are 11.579 and 12.244 over 6,193 held-out trials. They correspond to trial-pooled mean squared errors of 0.001870 and 0.001977, giving the same relative
745 reduction of 5.43%. All five frozen acceptance conditions pass, so the system authorizes only bounded population-response prediction within this resource. A later descriptive range-read audit verifies that finite contrast and direction counts match the frozen trial counts for all 32 mice. Every held-out contrast–direction combination and every held-out trial condition
750 occurs in the corresponding training partition. No mouse has a test-only contrast, direction, or contrast–direction combination. This audit does not alter the split, model, thresholds, or authorization.

The one-at-a-time sensitivity results preserve the positive decision at the

Table 11: Post-outcome, one-at-a-time sensitivity of the author-defined DANDI acceptance criteria. The frozen threshold is the prospective decision rule. The last column reports the nearest tested decision boundary or, when no boundary is crossed, the observed outcome. It is not a recommended replacement or a criterion-valid standard.

Criterion	Frozen	Observed	Sensitivity boundary or outcome
DANDI:000039 usable mice	≥ 25	32	≥ 33
Median mouse correlation	≥ 0.10	0.310	≥ 0.35
Bootstrap lower bound	> 0	0.207	> 0.25
Positive-mouse fraction	≥ 0.65	0.938	≥ 0.95
Relative squared-error gain	> 0	5.43%	$> 7.5\%$
DANDI:001176 usable subjects	≥ 20	5	Negative at every tested minimum ≥ 10 and available only at ≤ 5

755 frozen DANDI:000039 thresholds and expose finite margins. Within the
declared grids, the decision fails when the subject requirement exceeds 32,
the median-correlation threshold reaches 0.35, the bootstrap-lower-bound re-
quirement reaches 0.25, the positive-subject requirement reaches 0.95, or a
minimum 7.5% relative squared-error improvement is added. DANDI:001176
remains unavailable for every tested requirement of at least 10 subjects. It
760 would pass availability only if the minimum were relaxed to the observed
five subjects, which is exactly the outcome-informed repair prohibited by the
frozen protocol. These post-outcome checks describe sensitivity around two
realized outcomes. They provide no criterion validity for the original values.

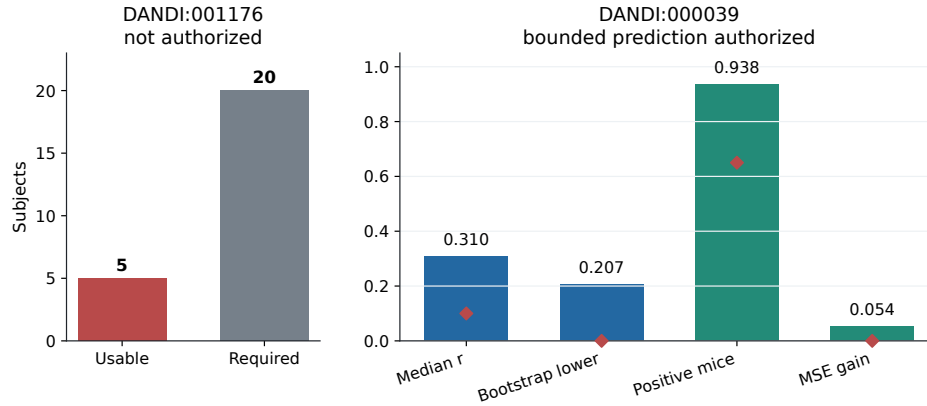


Figure 5: Prospective DANDI outcomes under rules fixed before numerical asset access. DANDI:001176 provides 5 usable subjects against the predeclared minimum of 20, so no model is fitted. DANDI:000039 passes every frozen condition across 32 mice, with median held-out correlation 0.310, subject-bootstrap lower bound 0.207, 93.8% positive mice, and relative MSE reduction 0.054. Red diamonds mark the respective thresholds of 0.10, 0, 0.65, and 0. Bar heights are criterion-specific values and are not intended for direct cross-metric comparison. Only bounded population-response prediction is authorized.

The positive result is not a state-of-the-art model claim. Ridge regression
 765 was chosen because the experiment tests prospective authorization rather
 than neural architecture. The negative result is equally part of the method
 evaluation. Together they show that a frozen knowledge contract can pre-
 serve both authorization and refusal without post-outcome threshold repair.
 The sensitivity analysis was executed later and cannot alter either decision.
 770 The final release verifies all source revisions, profile and protocol hashes, exact
 public DANDI asset identifiers, and the absence of `-dirty` artifacts.

Table 12: Artifact-grounded and prospective mouse-brain decisions. Authorization never extends beyond the claim stated in the second column.

Resource	Bounded target claim	Decision	Decisive result or deficit
Allen VBN	Directed topology and prediction	No	Topology and direction fail, with further missing blocks
Static Sensorium	Bounded predictive performance	Yes	Five-dataset comparison passes declared prediction and quality rules
Dynamic Sensorium	Bounded predictive performance	No	Response reliability requires review
IBL	Topology-specific behavioural prediction	Yes	Locked two-split comparison passes, without exclusive superiority
MICRONS	Local structure-function association	Yes	Three windows pass dyadic covariance and node-label permutation
DANDI:001176	Cortical ACh prediction	No	5 usable subjects against a frozen minimum of 20
DANDI:000039	Population contrast-response prediction	Yes	32 mice, median $r = 0.310$, bootstrap lower bound 0.207
Strict digital twin	Entity-specific whole-brain twin	No	Zero authorizations across all evaluated resources

5. Discussion

The results support MouseClaimBench as a bounded knowledge-based authorization layer. They do not support a claim that the system deter-

775 mines scientific truth. Its principal function is to prevent a valid result in one evidential dimension from being promoted into a stronger conclusion that requires different evidence. This distinction is operational rather than rhetorical because authorization is computed from named requirements, package integrity, and category-complete deficit traces.

780 *5.1. Scientific interpretation and methodological novelty*

The standards experiment provides an important negative baseline. A third-party SHACL engine reproduces every structural decision and deficit. MouseClaimBench therefore cannot claim novelty for checking required graph fields, representing provenance, or exporting JSON-LD. These capabilities
785 are established by SHACL, PROV-O, and the broader semantic-web ecosystem. Similarly, assurance cases already represent claims and evidence, while scientific verification already retrieves and classifies documentary support.

The defensible contribution is the authorization-contract architecture that remains after established components are removed from the novelty
790 claim. It binds domain evidence semantics, structural conformance, provenance integrity, and category-complete non-compensatory deficits without merging their responsibilities. The profile distinguishes prediction, reproduction, topology, direction, intervention, local structure–function association, and digital-twin scope. The admissibility rule rejects incomplete passing
795 facts, while the integrity layer evaluates relations among artifacts, cohorts, attestations, and the profile itself. Python and the separately implemented ASP path agree on domain outputs across all 5,677 generated cases. External SHACL reproduces the declared structural outcomes while allowing scientifically failed but well-formed packages to conform. The dedicated final

800 object executes $S \wedge A \wedge I$ and retains deficits by layer. The knowledge ledger makes every author-policy relation and predicate boundary inspectable. One-edge perturbations identify decisive relations in fixed packages, while counterfactual tests establish that generated explanations are faithful to the declared semantics. Controlled attacks and trust-boundary controls
805 distinguish detectable inconsistency from unauthenticated but internally coherent metadata. Finally, frozen external protocols demonstrate that the system can retain a negative availability result and authorize a separate positive result without modifying the contract.

The one-edge perturbation result is asymmetric and deliberately limited.
810 Fifty-nine of the 60 relation removals do not change an observed authorization because their associated cases contain additional deficits. This does not make those requirements redundant in the profile. It only means that they are not singly decisive in the fixed cases. The remaining removal identifies data quality as the sole barrier in Dynamic Sensorium. Conversely, adding
815 an unprovided mandatory block can only contract an authorization under the non-compensatory rule. The 71 contractions therefore demonstrate expected monotonic behaviour and locate affected packages. They do not show that the added relations are scientifically appropriate. Threshold changes, mandatory-to-review substitutions, and independently proposed profiles re-
820 main distinct forms of sensitivity that the present one-edge experiment does not cover.

The explanation study supports a narrower statement than human usefulness. Full repair is sufficient, every repaired deficit category is individually necessary, and each detected category retains an actionable witness. The

825 reported 41,641 omitted deficits are the arithmetic difference between 51,480
recorded deficits and the 9,839 retained by a one-reason baseline. These are
formal information and counterfactual properties. Whether the additional in-
formation changes reviewer accuracy, time, confidence, or agreement remains
unknown.

830 The attack benchmark is stronger than a demonstration with a few hand-
selected failures because it covers every composition of the eight original
attack families for every claim. It is still narrower than a general security
evaluation. The zero false authorizations across 2,550 attacked packages and
exact invariant traces are conditional on those attacks. The extended regres-
835 sion suite closes dangling attestation, independence, cohort, unknown-block,
status-consistency, and missing-attestation cases, but it remains specification-
driven. More importantly, all 20 coherent-forgery controls escape. An at-
tacker who changes content and its asserted hash together or supplies false
but internally consistent independence metadata leaves no contradiction for
840 the gate to observe. The result therefore documents both controlled robust-
ness and a concrete failure boundary. Source trust, digital signatures, archive
compromise, and unmodelled dependence relations remain outside the cur-
rent gate.

The prospective applications strengthen temporal validity but should not
845 be confused with independent discovery. The DANDI:000039 contrast re-
sponse is biologically expected and the Ridge model is intentionally simple.
Its role is to test whether a precommitted data selection, endpoint, compara-
tor, uncertainty calculation, and authorization rule can survive contact with
an unseen asset catalog. The acceptance values have precommitment valid-

ity but no established criterion validity. The post-outcome sensitivity table
makes their finite decision margins explicit and does not legitimize threshold
replacement. DANDI:001176 reveals the complementary failure mode. Meta-
data listed 29 subjects, but only 5 supplied the simultaneous schema required
by the frozen endpoint. Lowering the minimum to five after this observation
would produce a more convenient experiment and weaker evidence.

The MICRONS result is also deliberately narrow. Directed dyadic co-
variance increases the discovery standard error from the naive HC1 value
by a factor of 1.785, which confirms that pair-level independence would be
anti-conservative. Discovery fixes the positive direction and its one-sided
permutation value is descriptive. Both pre-fixed hold-outs confirm that di-
rection under the dyadic and node-label permutation criteria. This supports
the declared local observational endpoint. It does not establish a synaptic
mechanism, external replication, or a complete visual-cortex model.

Lessons learned. Four practical lessons follow. Structural conformance
should be delegated to a standard engine whenever possible. Provenance
must include dependence and independence claims, not only file locations.
Negative outcomes should remain first-class release artifacts because they
expose unavailable evidence and prevent silent protocol repair. Finally,
a category-complete deficit set retains substantially more policy-relevant
information than one decisive label when multiple requirements fail simul-
taneously. Completeness applies to violated dimensions and their witnesses,
not to every redundant occurrence of the same violation. Its practical value
to human reviewers has not been measured.

5.2. Limitations and threats to validity

875 Construct validity remains the principal limitation. Profile v2 is author-defined and literature-grounded. Its 10 claims, 22 predicate contracts, 60 direct relations, claim-specific rationales, exceptions, and sources are now fully inspectable. However, no independent expert panel or community process has established that the claims, predicates, or relations are necessary, 880 sufficient, or complete. The frozen v2 external-review protocol contains 92 items but has zero completed independent raters. The one-edge perturbation cannot substitute for that assessment because many contractions follow directly from monotonic non-compensation. Soundness, completeness, counterfactual fidelity, and perturbation results are relative to this profile. They 885 cannot convert an author-defined policy into a consensus scientific taxonomy.

Internal validity is stronger for software behaviour than for scientific content. Expected outcomes in the mutation, counterfactual, and declared-attack benchmarks follow directly from the authorization semantics. Multiple implementations reduce the risk of a coding error, but Python, ASP, and 890 SHACL still share the same profile specification. The 221 one-edge variants expose local monotonic behaviour and do not span the space of scientifically plausible profiles. The DANDI threshold analysis is post-outcome and descriptive. It does not establish criterion validity. The attack generator is deterministic and non-adaptive, and the coherent-forgery controls confirm 895 that self-consistent false metadata remains unobservable. Property checks cover sampled packages rather than every possible future profile and package graph.

The real applications are heterogeneous and should not be pooled as

independent replications. Allen, Sensorium, IBL, MICRONS, and DANDI
900 answer different questions. The first five-package mapping reuses artifacts
from the preceding MouseBrainBench study [23] and provides retrospective
contract application rather than new data validation. The MICRONS win-
dows occupy one cortical volume. The two DANDI studies use different
measurements and cannot replicate one another. DANDI:000039 validates
905 a bounded predictive contract with a simple comparator, not model supe-
riority. DANDI:001176 is an availability failure rather than an estimate of
predictive performance.

Statistical uncertainty is handled at the declared independent unit where
possible. The DANDI bootstrap resamples mice, and MICRONS uses di-
910 rected dyadic covariance plus node-label permutation. These methods still
depend on their exchangeability and model assumptions. The MICRONS
hold-outs were fixed after discovery rather than externally preregistered, and
all three use the same resource. The DANDI protocols were frozen in Git
before numerical asset access but were not registered in a third-party registry.

915 External validity is not established outside computational mouse-brain
evidence. The architecture can load another profile, but architectural ex-
tensibility is not empirical generality. Human comprehensibility, decision
quality, review time, and inter-rater agreement were not evaluated. The per-
formance experiment comprises component microbenchmarks on one Apple
920 arm64 computer. It excludes archive download, source-predicate execution,
graph serialization, and external SHACL validation. These restrictions are
encoded in the release gate and should remain visible in any subsequent use.

6. Conclusions and Future Work

MouseClaimBench provides an integrity-aware knowledge-based method
925 for authorizing bounded scientific claims from computational mouse-brain
evidence. Its authorization contracts jointly bind graph conformance,
package integrity, domain evidence semantics, and category-complete non-
compensatory deficits. Authorization requires structural conformance, every
mandatory domain block to be admissibly passed, and an integrity-clean
930 package. The result is tied to a named profile and includes every violated
deficit category, an actionable witness for each category, artifact identities,
and protocol revisions.

The revised evaluation separates acquisition, specification verification,
controlled threats, and empirical application. The acquisition ledger covers
935 all 60 claim–evidence relations and 22 predicate contracts without claiming
external consensus. Python and the separately implemented ASP path agree
on domain decisions and deficits across all 5,677 generated cases. External
SHACL independently recovers the expected structural decisions and struc-
tural deficits, including the canonical $S = 1$, $A = 0$ case. The executable
940 final gate reproduces all eight states of $S \wedge A \wedge I$. A one-edge perturbation
evaluates 221 profiles and 3,094 fixed artifact decisions. Counterfactual re-
pair and minimality checks show no violation across 9,839 repaired packages
and 51,480 individual deficits. These results establish implementation con-
formance and explanation fidelity relative to profile v2. They do not validate
945 the scientific completeness of the policy.

The integrity results cover 2,550 attacked packages spanning all 255
non-empty compositions of eight original attack families across each of 10

claims, with exact invariant traces and no false authorization. A separate 100-package suite verifies nine additional reference and attestation regressions. Six direct API regressions add duplicate-identifier, reflexive-relation, and composite data-generation identity cases. The 20 coherent-forgery controls all remain authorized, which exposes rather than conceals the unresolved need for external trust anchors. The scientific applications preserve the same boundaries. One prospective DANDI protocol remains negative because its frozen subject requirement is unavailable. A second authorizes only population-response prediction across 32 mice. Post-outcome threshold sensitivity exposes both decisions without providing criterion validity. Dependence-aware MICRONS inference confirms one local observational structure–function association in two pre-fixed hold-outs. No evaluated case authorizes a causal mechanism, external biological replication, whole-brain coverage, or a complete entity-specific mouse-brain digital twin.

The next scientific requirement is independent assessment of the 92 v2 claim, predicate, and relation items by real domain experts, followed by evaluation of category-complete traces with human users. Independently proposed profiles, mandatory-to-review substitutions, and externally justified thresholds are needed for stronger sensitivity and construct-validity evidence. The integrity model should add signed attestations, archive trust policies, and adaptive attacks, while future external endpoints should be registered across laboratories and animals. The immediate conclusion remains narrower and testable: predictive accuracy, reproducible execution, and valid graph structure cannot compensate for missing scientific evidence or incoherent provenance under the declared profile.

Declaration of Competing Interest

The authors declare no known competing financial interests or personal
975 relationships that could have appeared to influence the work reported in this
paper.

CRedit Authorship Contribution Statement

Alberto Fernández-Isabel: Conceptualisation, Methodology, Software, Inves-
tigation, Data curation, Writing–original draft, Writing–review and editing.

980 Isaac Martín de Diego: Validation, Formal analysis, Investigation, Writing–
original draft, Writing–review and editing, Project administration.

Rubén Fuentes-Fernández: Methodology, Validation, Writing–original draft,
Writing–review and editing.

Data and Code Availability

985 The source code, knowledge profiles, tests, frozen protocols, lightweight
result artifacts, and manuscript are maintained in the public MouseClaim-
Bench repository at <https://github.com/Fiutten/MouseClaimBench>.
The software snapshot used for this revision is released as tag v0.12.2 at
<https://github.com/Fiutten/MouseClaimBench/tree/v0.12.2>. Each
990 regenerated revision artifact records the clean source commit used for its
execution. Raw Allen, Sensorium, IBL, MICRONS, and DANDI data re-
main governed by their providers and are not redistributed. The prospective
artifact manifests retain public identifiers, versions, sizes, and official hashes.
Every reported result is linked to a recorded Git revision, and artifacts
995 marked as dirty are excluded from the release gate. A persistent DOI has

not been assigned, so the repository and tagged commit are reported without implying DOI-backed archival permanence.

Declaration of Generative AI and AI-Assisted Technologies

During preparation of this work, the authors used OpenAI Codex, an
1000 Artificial Intelligence (AI)-assisted tool, for software assistance and language editing. The authors reviewed and edited all generated material and take full responsibility for the content of the publication.

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